DIOPHANTINE SINGULARITY AND CONDITIONING OF SHIFTED KAC–MURDOCK–SZEGŐ MATRICES

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ABSTRACT. For $N \geq 2$, let $T_N(a) = [a^{|i-j|}]_{i,j=0}^{N-1}$, with $0 < a < 1$, be the $N \times N$ Kac–Murdock–Szegő matrix, and fix a real level τ. As the dimension varies, we ask when τ is an eigenvalue and how ill-conditioned $T_N(a) - \tau I_N$ becomes when the spectrum nearly hits it. Outside the range of the KMS symbol the condition numbers stay uniformly bounded, while at either band edge they grow quadratically with explicit constants. Inside the band, the exact KMS phase law converts the problem to one inhomogeneous Diophantine residual $d_N = \operatorname{dist}(N\alpha - \beta, \mathbb{Z})$ in smooth coordinates (α, β) on an open triangle.

Exact singularity is equivalent to $d_N = 0$. Hence the number of singular dimensions is always 0, 1, or ∞; all three behaviors are dense, although the never-singular parameters have full Lebesgue measure. At every sufficiently large nonsingular interior dimension, $\sigma_{\min}(T_N(a) - \tau I_N)$ is an explicit constant times $d_N/(N + \tau)$, so the condition number is an explicit constant times $(N + \tau)/d_N$. Rational α gives periodic behavior. Irrational α gives an equidistributed bulk, a Pareto condition-number law of tail index 1, and rare spikes controlled by one-sided inhomogeneous approximation. The hollow case $\tau = 1$ is the resonant diagonal $\beta = \alpha$, where the arithmetic becomes homogeneous.

1. INTRODUCTION

Fix $0 < a < 1$ and consider the real symmetric Kac–Murdock–Szegő matrix

$$T_N(a) = \left[a^{|i-j|}\right]_{i,j=0}^{N-1}, \qquad N \geq 2. \tag{1}$$

These matrices are classical finite AR(1) correlation matrices. Fix a real spectral level τ and consider the shifted family

$$B_N(a, \tau) = T_N(a) - \tau I_N. \tag{2}$$

Here I_N is the $N \times N$ identity matrix. Our question is cross-dimensional: as N varies, when is τ an eigenvalue of $T_N(a)$, and how close does the spectrum come when it is not? Because B_N is symmetric, this is also a conditioning problem. Its smallest singular value is the distance from τ to the spectrum of $T_N(a)$. Thus an exact hit makes B_N singular, while a near hit makes it ill-conditioned. At nonsingular dimensions we write

$$\kappa_2(B_N) = \frac{\sigma_{\max}(B_N)}{\sigma_{\min}(B_N)}$$

for the spectral condition number.

The KMS symbol is the Poisson kernel

$$\mu_a(\theta) = \frac{1 - a^2}{1 - 2a\cos\theta + a^2}, \qquad 0 \leq \theta \leq \pi, \tag{3}$$

strictly decreasing on $(0, \pi)$ with range

$$\mu_-(a) = \frac{1 - a}{1 + a}, \qquad \mu_+(a) = \frac{1 + a}{1 - a}. \tag{4}$$

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These three positions lead to three different mechanisms. Outside the symbol band there is a fixed spectral gap. At a band edge the symbol meets the level quadratically. In the interior there is a genuine crossing, and it is here that the arithmetic enters.

The mechanism can be stated before it is proved. The finite KMS eigenangles obey an exact monotone phase law

$$N\theta + \psi_a(\theta) = k\pi, \qquad \psi_a'(\theta) = \mu_a(\theta).$$

For an interior level $\mu_-(a) < \tau < \mu_+(a)$, let θ_τ denote the unique crossing angle defined by

$$\mu_a(\theta_\tau) = \tau. \tag{5}$$

Set

$$\alpha = \frac{\theta_\tau}{\pi}, \qquad \beta = 1 - \frac{\psi_a(\theta_\tau)}{\pi}. \tag{6}$$

At the crossing, the phase location is $N\alpha + 1 - \beta$. Its distance to the nearest integer is therefore

$$d_N = \operatorname{dist}(N\alpha - \beta, \mathbb{Z}), \tag{7}$$

Sections 2–3 show that (α, β) are exact coordinates on the interior parameter region and that this same residual appears in the determinant. In particular, B_N is singular exactly when $d_N = 0$. The cross-dimensional spectral problem is therefore reduced to a single arithmetic question: how closely does the orbit $N\alpha$ approach the target β modulo one?

To state the conditioning law, define

$$\gamma(a,\tau) = |\mu_a'(\theta_\tau)|, \quad M(a,\tau) = \max\{\mu_+ - \tau, \tau - \mu_-\}, \quad C(a,\tau) = \frac{M(a,\tau)}{\pi\gamma(a,\tau)}. \tag{8}$$

Theorem 1 (Main conditioning theorem)**.** *Fix $0 < a < 1$ and let $N \to \infty$.*

(1) *If $\tau < \mu_-$, then*

$$\kappa_2(B_N) < \frac{\mu_+ - \tau}{\mu_- - \tau}, \qquad \kappa_2(B_N) \longrightarrow \frac{\mu_+ - \tau}{\mu_- - \tau}.$$

(2) *If $\tau = \mu_-$, then*

$$\kappa_2(B_N) = \frac{4\mu_+^2}{\pi^2}(N + \mu_-)^2\big(1 + O(N^{-2})\big).$$

(3) *If $\mu_- < \tau < \mu_+$, then at every sufficiently large nonsingular N,*

$$\sigma_{\min}(B_N) = \pi\gamma(a,\tau)\frac{d_N}{N+\tau}\big(1 + O(N^{-1})\big), \qquad \kappa_2(B_N) = C(a,\tau)\frac{N+\tau}{d_N}\big(1 + O(N^{-1})\big).$$

(4) *If $\tau = \mu_+$, then*

$$\kappa_2(B_N) = \frac{4\mu_-^2}{\pi^2}(N + \mu_+)^2\big(1 + O(N^{-2})\big).$$

(5) *If $\tau > \mu_+$, then*

$$\kappa_2(B_N) < \frac{\tau - \mu_-}{\tau - \mu_+}, \qquad \kappa_2(B_N) \longrightarrow \frac{\tau - \mu_-}{\tau - \mu_+}.$$

The interior error terms are for fixed (a,τ); they are not uniform as τ approaches a band edge.

The theorem cleanly separates analysis from arithmetic. Outside the band, conditioning stays bounded; at either edge it grows quadratically. In the interior, all irregularity is concentrated in the scalar sequence d_N. For rational α, this sequence is periodic, and exact singularity is either absent or recurrent on one residue class. For irrational α, at most one exact singular dimension can occur; equidistribution governs the bulk, while unusually small values of d_N produce exceptional conditioning spikes. Consequently the number of singular dimensions is always 0, 1, or ∞.

The special shift $\tau = 1$, for which $B_N = T_N - I_N$ has zero diagonal, is not a separate mechanism. In the coordinates above it is the diagonal $\beta = \alpha$, the $N = 1$ member of the same affine singularity

web. There the residual becomes $\operatorname{dist}((N-1)\alpha, \mathbb{Z})$, so the inhomogeneous problem reduces to homogeneous approximation. We return to this hollow resonance only after the general shifted problem is in place.

The classical inputs and the contribution claimed here should be kept distinct. KMS spectral formulas, characteristic-polynomial recurrences, Toeplitz phase methods, and the Diophantine approximation theorems used later all have substantial prior literature [1, 2, 4, 3, 7, 8, 19, 9]. What is claimed here is the KMS-specific fixed-level dictionary: explicit interior coordinates, the affine residual $N\alpha - \beta$, an exact singularity classification, a determinant formula organized around that residual, and the resulting conditioning laws with explicit leading constants. Section 8 states the detailed prior-art boundary.

The paper follows that dictionary in dependency order. Section 2 puts the classical KMS spectrum into the exact monotone phase form needed here. Section 3 converts a fixed interior level into the coordinates (α, β) and derives the determinant formula. Section 4 classifies exact singularity and recurrence. Section 5 proves the five conditioning regimes. Section 6 then reads the interior formula through rational periodicity, irrational equidistribution, and one-sided inhomogeneous approximation. Section 7 recovers the hollow case as a resonant slice of the same geometry.

2. KMS quantization and Blaschke phase

Two spectral facts drive everything that follows: the eigenangles are level sets of one strictly increasing phase, and the derivative of that phase is $N + \mu_a(\theta)$. This section derives both facts from the classical KMS recurrence. It also records the disk-automorphism form of the boundary phase, which makes the identity $\psi_a' = \mu_a$ transparent.

Throughout the rest of the paper we use

$$\mu_a(\theta) = \frac{1-a^2}{1-2a\cos\theta + a^2}, \qquad \mu_- = \frac{1-a}{1+a}, \qquad \mu_+ = \frac{1+a}{1-a}.$$

Thus $\mu_- \le \mu_a(\theta) \le \mu_+$ on $[0, \pi]$.

The inverse of (1) is

$$T_N(a)^{-1} = \frac{1}{1-a^2}\begin{pmatrix} 1 & -a & & & \\ -a & 1+a^2 & -a & & \\ & \ddots & \ddots & \ddots & \\ & & -a & 1+a^2 & -a \\ & & & -a & 1 \end{pmatrix}; \tag{9}$$

see [1]. The corresponding second-order recurrence gives the standard secular function

$$S_N(\theta) = \sin((N+1)\theta) - 2a\sin(N\theta) + a^2\sin((N-1)\theta). \tag{10}$$

Define

$$D(\theta) = (1+a^2)\cos\theta - 2a, \qquad E(\theta) = (1-a^2)\sin\theta, \tag{11}$$

and, for $0 < \theta < \pi$,

$$\psi_a(\theta) = \arg(D(\theta) + iE(\theta)) \in (0, \pi). \tag{12}$$

Put

$$R(\theta) = \sqrt{D(\theta)^2 + E(\theta)^2} = 1 - 2a\cos\theta + a^2.$$

Lemma 2 (Phase form)**.** *For $0 < \theta < \pi$,*

$$S_N(\theta) = R(\theta)\sin\big(N\theta + \psi_a(\theta)\big). \tag{13}$$

Moreover ψ_a extends continuously to $[0, \pi]$ with $\psi_a(0) = 0$ and $\psi_a(\pi) = \pi$.

Proof. Expanding the outer sine terms in (10) gives

$$S_N(\theta) = D(\theta)\sin(N\theta) + E(\theta)\cos(N\theta),$$

which is (13). At the endpoints E vanishes while $D(0) = (1-a)^2 > 0$ and $D(\pi) = -(1+a)^2 < 0$. □

Proposition 3 (Disk-automorphism form of the phase)**.** *Let*

$$b_a(z) = \frac{z-a}{1-az}. \tag{14}$$

For $0 < \theta < \pi$*,*

$$\boxed{e^{i\psi_a(\theta)} = b_a(e^{i\theta}).} \tag{15}$$

Consequently

$$\boxed{\psi_a'(\theta) = \frac{1-a^2}{|1-ae^{i\theta}|^2} = \mu_a(\theta).} \tag{16}$$

At every eigenangle θ_k*, with* $z_k = e^{i\theta_k}$*,*

$$\boxed{z_k^N b_a(z_k) = (-1)^k.} \tag{17}$$

Proof. There is an exact factorization

$$D(\theta) + iE(\theta) = e^{-i\theta}(e^{i\theta} - a)^2.$$

Since

$$|e^{i\theta} - a|^2 = (e^{i\theta} - a)(e^{-i\theta} - a),$$

dividing by the modulus gives

$$\frac{D + iE}{|D + iE|} = \frac{e^{i\theta} - a}{1 - ae^{i\theta}},$$

which is (15). The angular derivative of the boundary argument of the disk automorphism b_a is the Poisson kernel,

$$\frac{d}{d\theta}\arg b_a(e^{i\theta}) = \frac{1-a^2}{|1-ae^{i\theta}|^2},$$

giving (16). Finally, the exact phase quantization below is $N\theta_k + \psi_a(\theta_k) = k\pi$; exponentiation gives (17). □

Remark 4. The use of Blaschke boundary phases and Prüfer-type arguments in zero-spacing problems is classical; see, for example, Simanek [21]. Likewise, finite Bernstein–Szegő Jacobi systems admit exact phase equations built from finite products of Möbius factors [22]. Proposition 3 is recorded because in the KMS family the same Poisson kernel is simultaneously the spectral symbol, which is what later turns the generic local factor $N + \psi_a'(\theta_\tau)$ into $N + \tau$.

Set

$$G_N(\theta) = N\theta + \psi_a(\theta). \tag{18}$$

Then

$$G_N'(\theta) = N + \mu_a(\theta) > 0. \tag{19}$$

Theorem 5 (Exact quantization)**.** *The* N *KMS eigenangles are the unique points*

$$0 < \theta_1 < \cdots < \theta_N < \pi$$

satisfying

$$G_N(\theta_k) = k\pi, \qquad k = 1, \ldots, N. \tag{20}$$

They are simple and obey

$$\frac{\pi}{N + \mu_+} \le \theta_{k+1} - \theta_k \le \frac{\pi}{N + \mu_-}. \tag{21}$$

The corresponding KMS eigenvalues and eigenvectors are

$$\mu_k = \mu_a(\theta_k), \qquad v_j^{(k)} = \sin((j+1)\theta_k) - a\sin(j\theta_k), \quad j = 0, \dots, N-1. \tag{22}$$

Hence the eigenvalues of $B_N(a,\tau)$ are

$$\lambda_k(a,\tau) = \mu_a(\theta_k) - \tau. \tag{23}$$

Proof. By Lemma 2, the roots of the secular equation are the preimages of $\pi\mathbb{Z}$ under G_N. Equation (19) shows that G_N increases from 0 to $(N+1)\pi$, giving exactly (20). The spacing estimate follows from the mean-value theorem and $\mu_- \le \mu_a \le \mu_+$. The eigenvalue and eigenvector formulas are the standard solution of the recurrence from (9); see also [3, 7, 8]. Subtracting τI_N shifts the eigenvalues without changing the eigenvectors. □

3. The Diophantine triangle and the determinant

The phase law locates every finite eigenangle. For a fixed spectral level, we now freeze the crossing angle and record its position relative to that phase grid. Two steps are needed: identify the original parameter region in coordinates (α, β), then show that the same affine residual $N\alpha - \beta$ appears in the characteristic polynomial.

Assume throughout this section that

$$\mu_-(a) < \tau < \mu_+(a). \tag{24}$$

There is a unique $\theta_\tau \in (0,\pi)$ satisfying (5), with

$$\cos\theta_\tau = \frac{1 + a^2 - (1-a^2)/\tau}{2a}. \tag{25}$$

Define

$$\alpha = \frac{\theta_\tau}{\pi}, \qquad \beta = 1 - \frac{\psi_a(\theta_\tau)}{\pi}. \tag{26}$$

Theorem 6 (Diophantine-triangle coordinates)**.** *The map $(a,\tau) \mapsto (\alpha,\beta)$ is a bijection from the interior parameter region*

$$\{(a,\tau) : 0 < a < 1,\ \mu_-(a) < \tau < \mu_+(a)\}$$

onto the open triangle

$$\mathcal{D} = \{(\alpha,\beta) : \alpha > 0,\ \beta > 0,\ \alpha + \beta < 1\}. \tag{27}$$

The inverse is

$$\boxed{\tau = \frac{\sin(\pi\beta)}{\sin(\pi\alpha)}, \qquad a = \frac{\cos\frac{\pi}{2}(\alpha+\beta)}{\cos\frac{\pi}{2}(\alpha-\beta)}.} \tag{28}$$

Proof. For a general interior angle θ, write $\psi = \psi_a(\theta)$ and $D + iE = Re^{i\psi}$. Then

$$R\sin(\psi - \theta) = E\cos\theta - D\sin\theta = 2a(1 - a\cos\theta)\sin\theta > 0.$$

Since $0 < \theta, \psi < \pi$, this gives

$$0 < \theta < \psi < \pi.$$

At the crossing, $\theta = \pi\alpha$ and $\psi = \pi(1-\beta)$, hence $\alpha > 0$, $\beta > 0$ and $\alpha + \beta < 1$.

Next, $E = R\sin\psi = (1-a^2)\sin\theta$, while $\tau = (1-a^2)/R$ at the crossing. Thus

$$\tau = \frac{\sin\psi}{\sin\theta} = \frac{\sin(\pi\beta)}{\sin(\pi\alpha)}.$$

For the formula for a, observe the exact factorization

$$D + iE = e^{-i\theta}(e^{i\theta} - a)^2. \tag{29}$$

If $\gamma = \arg(e^{i\theta} - a)$, then $\psi = 2\gamma - \theta$ and therefore $\gamma = (\psi + \theta)/2$. Resolving the triangle formed by $e^{i\theta} - a$ gives

$$a = \frac{\sin\frac{1}{2}(\psi - \theta)}{\sin\frac{1}{2}(\psi + \theta)} = \frac{\cos\frac{\pi}{2}(\alpha + \beta)}{\cos\frac{\pi}{2}(\alpha - \beta)}.$$

Conversely, let (α, β) satisfy (27) and put $\theta = \pi\alpha$, $\psi = \pi(1 - \beta)$ and $\gamma = (\theta + \psi)/2$. Then $0 < \theta < \psi < \pi$. Formula (28) gives

$$a = \frac{\sin(\gamma - \theta)}{\sin\gamma},$$

so $0 < a < 1$ (equivalently, in the cosine form of (28), the numerator and denominator are positive and the numerator is smaller). Moreover

$$e^{i\theta} - a = \frac{\sin\theta}{\sin\gamma} e^{i\gamma}.$$

Using (29), the corresponding boundary phase is therefore $2\gamma - \theta = \psi$. Since $E = R\sin\psi = (1 - a^2)\sin\theta$, we obtain

$$\mu_a(\theta) = \frac{1 - a^2}{R} = \frac{\sin\psi}{\sin\theta} = \frac{\sin(\pi\beta)}{\sin(\pi\alpha)} = \tau.$$

Thus θ is the crossing angle and the constructed pair maps back to (α, β), proving surjectivity and uniqueness. □

The coordinate theorem locates the fixed level in phase space. Exact singularity, however, is a statement about the characteristic polynomial at that same level. The following shifted Chebyshev form supplies the bridge. It is recorded here in a self-contained form and is equivalent to classical KMS characteristic-polynomial formulas; see [3].

The determinant recurrence itself is classical. Chremmos and Fikioris derive a second-order recurrence for the characteristic polynomial of the same geometric Toeplitz family [4]; under the real KMS specialization, that machinery contains the hollow ($\tau = 1$) determinant identity. We use the inverse/Chebyshev route below because it places the general scalar shift directly in the (a, τ) variables needed for the fixed-level problem.

Lemma 7 (Shifted Chebyshev determinant)**.** *For $N \geq 2$, $0 < a < 1$, and $\tau \neq 0$,*

$$\det(T_N(a) - \tau I_N) = \frac{(\tau a)^N}{1 - a^2}\big[U_N(x) + 2aU_{N-1}(x) + a^2 U_{N-2}(x)\big], \tag{30}$$

where U_j is the Chebyshev polynomial of the second kind and

$$x = \frac{1 - a^2 - \tau(1 + a^2)}{2\tau a}. \tag{31}$$

Proof. The classical KMS inverse is

$$T_N(a)^{-1} = \frac{1}{1 - a^2}\begin{pmatrix} 1 & -a & & & \\ -a & 1 + a^2 & -a & & \\ & \ddots & \ddots & \ddots & \\ & & -a & 1 + a^2 & -a \\ & & & -a & 1 \end{pmatrix}, \qquad \det T_N(a) = (1 - a^2)^{N-1}.$$

Hence

$$\det(T_N - \tau I_N) = \frac{\det H_N}{1 - a^2},$$

where H_N is tridiagonal with off-diagonal $b = \tau a$, interior diagonal $c = 1 - a^2 - \tau(1 + a^2) = 2bx$, and corner diagonal $d = 1 - a^2 - \tau = b(2x + a)$. Let P_k be the determinant of the leading $k \times k$ block before the final corner modification. Then

$$P_0 = 1, \qquad P_1 = d, \qquad P_k = cP_{k-1} - b^2 P_{k-2},$$

so, with $U_{-1} = 0$,

$$P_k = b^k \big(U_k(x) + aU_{k-1}(x)\big).$$

Applying the last corner gives

$$\det H_N = dP_{N-1} - b^2 P_{N-2} = b^N \big(U_N(x) + 2aU_{N-1}(x) + a^2 U_{N-2}(x)\big),$$

by the Chebyshev recurrence $U_j = 2xU_{j-1} - U_{j-2}$. This proves (30). □

For $x \in \mathbb{R}$, write $\|x\| = \operatorname{dist}(x, \mathbb{Z})$ for distance to the nearest integer.

Theorem 8 (Exact determinant)**.** *For every interior pair (a, τ) and every $N \geq 2$,*

$$\boxed{\det B_N(a, \tau) = (-1)^{N+1} a^N \tau^{N-1} \frac{\sin \pi(N\alpha - \beta)}{\sin(\pi\alpha)}.} \tag{32}$$

Consequently, with

$$d_N = \|N\alpha - \beta\| = \operatorname{dist}(N\alpha - \beta, \mathbb{Z}), \tag{33}$$

$$|\det B_N(a, \tau)| = a^N \tau^{N-1} \frac{\sin(\pi d_N)}{\sin(\pi\alpha)}. \tag{34}$$

Proof. At the crossing, (25) implies $x = -\cos\theta_\tau$. Since $U_j(-\cos\theta) = (-1)^j \sin((j+1)\theta)/\sin\theta$, the bracket in (30) becomes

$$(-1)^N \frac{S_N(\theta_\tau)}{\sin\theta_\tau}.$$

By Lemma 2,

$$S_N(\theta_\tau) = R(\theta_\tau) \sin(N\theta_\tau + \psi_a(\theta_\tau)).$$

Since $R(\theta_\tau) = (1 - a^2)/\tau$, $\theta_\tau = \pi\alpha$, and $\psi_a(\theta_\tau) = \pi(1 - \beta)$, substitution into (30) yields (32). Taking absolute values and reducing the sine modulo integers gives (34). □

4. Exact singularity geometry and recurrence

The determinant formula reduces exact singularity to one affine arithmetic equation. We classify that equation before returning to conditioning. The decisive split is rational versus irrational α: rational phase points can recur periodically, whereas an irrational phase point can lie on the singularity web at most once.

Theorem 9 (Exact singularity)**.** *For an interior parameter pair,*

$$\boxed{B_N(a, \tau) \text{ is singular} \iff N\alpha - \beta \in \mathbb{Z}.} \tag{35}$$

At a singular dimension the nullity is exactly one.

Proof. The criterion follows from Theorem 8. Simplicity follows from the simple KMS spectrum in Theorem 5. □

The geometry of (35) is the family of affine lines

$$\beta = N\alpha - m, \qquad m \in \mathbb{Z}. \tag{36}$$

We refer to their intersections with $\mathcal{D}$ as the *affine singularity web*.

For rational α, the web condition becomes a finite congruence problem.

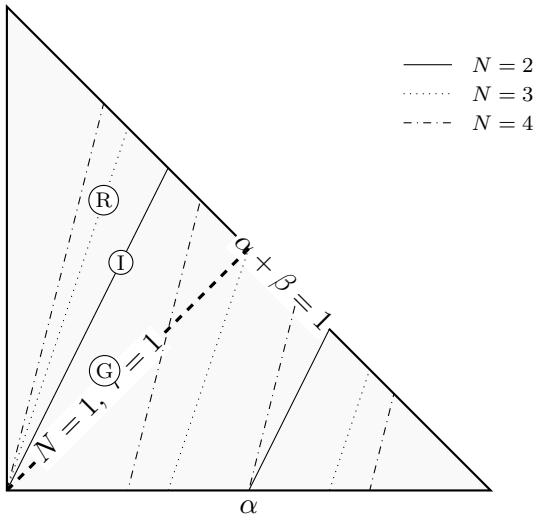


FIGURE 1. The Diophantine target triangle $\alpha > 0$, $\beta > 0$, $\alpha + \beta < 1$. Exact singularity at dimension N lies on the affine lines $\beta \equiv N\alpha \pmod 1$; representative lines for $N = 2, 3, 4$ are shown. The dashed diagonal is the $N = 1$ member of the web, just outside our nontrivial matrix range, and hence the resonant shift $\tau = 1$. The marked points illustrate a rational recurrent point (R), an irrational point with one exact physical hit (I), and a generic off-web point (G). Theorem 13 shows that all three behaviors are dense in the full triangle, although the off-web, never-singular stratum has full two-dimensional measure.

Theorem 10 (Rational recurrence)**.** *Let $\alpha = p/q$ in lowest terms. Then singularity occurs for some dimension if and only if*

$$q\beta \in \mathbb{Z}. \tag{37}$$

If $r = q\beta \in \mathbb{Z}$, then $0 < r < q - p$ and the singular dimensions are exactly the one congruence class

$$Np \equiv r \pmod q. \tag{38}$$

If $q\beta \notin \mathbb{Z}$, there are no singular dimensions.

Proof. The condition $Np/q - \beta \in \mathbb{Z}$ implies $Np - q\beta \in q\mathbb{Z}$, so a solution requires $q\beta \in \mathbb{Z}$. Conversely, if $r = q\beta$ is an integer, then the criterion is (38). Since p is invertible mod q, this determines exactly one residue class. The triangle inequalities imply $0 < r < q - p$. □

Corollary 11 (Complete recurrent parametrization)**.** *Every interior parameter pair with infinitely many singular dimensions is obtained from integers*

$$0 < p, \qquad 0 < r, \qquad p + r < q, \qquad \gcd(p, q) = 1,$$

by

$$\boxed{\alpha = \frac{p}{q}, \qquad \beta = \frac{r}{q}, \qquad \tau = \frac{\sin(r\pi/q)}{\sin(p\pi/q)}, \qquad a = \frac{\cos((p + r)\pi/(2q))}{\cos((p - r)\pi/(2q))}.} \tag{39}$$

The singular dimensions satisfy $Np \equiv r \pmod q$.

A recurrent non-hollow family. Take

$$\alpha = \frac{1}{5}, \qquad \beta = \frac{3}{5}.$$

Then $\alpha+\beta=4/5<1$, and Corollary 11 gives

$$a=\frac{\cos(2\pi/5)}{\cos(\pi/5)}=\frac{3-\sqrt{5}}{2},\qquad \tau=\frac{\sin(3\pi/5)}{\sin(\pi/5)}=\frac{1+\sqrt{5}}{2}.$$

The singularity congruence is

$$N\equiv 3 \pmod 5.$$

Thus the same non-hollow eigenvalue $\tau=\varphi$ recurs at every dimension $N\equiv 3 \pmod 5$ for the KMS decay parameter $a=(3-\sqrt{5})/2$.

The rational case permits periodic return. Irrational α behaves in the opposite way: two exact hits would force a rational relation.

Theorem 12 (Irrational uniqueness and the $0/1/\infty$ law)**.** *If α is irrational, $B_N(a,\tau)$ is singular for at most one dimension. Consequently, for every interior parameter pair,*

$$\boxed{\#\{N\geq 2 : B_N(a,\tau) \text{ singular}\}\in\{0,1,\infty\}.} \tag{40}$$

Proof. If two distinct dimensions N_1,N_2 were singular, then

$$N_1\alpha-\beta=m_1,\qquad N_2\alpha-\beta=m_2$$

for integers m_1,m_2. Subtraction gives $(N_1-N_2)\alpha=m_1-m_2\in\mathbb{Z}$, forcing $\alpha\in\mathbb{Q}$. The classification now follows from Theorem 10. □

Cardinality alone does not describe how these three behaviors sit in parameter space. The next theorem records both their topological abundance and their measure.

Theorem 13 (Dense singularity stratification)**.** *For $j\in\{0,1,\infty\}$ define*

$$\mathcal{S}_j=\{(\alpha,\beta)\in\mathcal{D}:\#\{N\geq 2 : B_N(a,\tau) \text{ is singular}\}=j\},$$

where (a,τ) is the parameter pair corresponding to (α,β) through Theorem 6. Then $\mathcal{S}_0$, $\mathcal{S}_1$ and $\mathcal{S}_\infty$ are all dense in $\mathcal{D}$. Moreover,

1. *$\mathcal{S}_0$ has full two-dimensional Lebesgue measure in $\mathcal{D}$;*
2. *$\mathcal{S}_1$ is uncountable and has two-dimensional measure zero;*
3. *$\mathcal{S}_\infty$ is countable and has two-dimensional measure zero.*

The same statements hold in the physical interior parameter region

$$\mathcal{P}=\{(a,\tau):0<a<1,\ \mu_-(a)<\tau<\mu_+(a)\}.$$

In particular, arbitrarily close to every interior shifted KMS matrix is one that is never singular, one that is singular exactly once, and one that is singular infinitely often, yet almost every interior parameter pair is never exactly singular.

Proof. Let

$$\mathcal{W}=\bigcup_{N\geq 2}\bigcup_{m\in\mathbb{Z}}\{(\alpha,\beta)\in\mathcal{D}:\beta=N\alpha-m\}$$

be the affine singularity web. It is a countable union of line segments and hence has two-dimensional measure zero. By Theorem 9, $\mathcal{S}_0=\mathcal{D}\backslash\mathcal{W}$, so $\mathcal{S}_0$ has full measure; it is dense because every nonempty open subset of $\mathcal{D}$ has positive area.

The recurrent set $\mathcal{S}_\infty$ is countable by Theorem 10. It is dense: given $(\alpha_0,\beta_0)\in\mathcal{D}$, choose arbitrarily large primes q and integers p,r with

$$\frac{p}{q}\to\alpha_0,\qquad \frac{r}{q}\to\beta_0,\qquad p+r<q.$$

For large q one has $1\leq p\leq q-1$, so $\gcd(p,q)=1$, while $q(r/q)=r\in\mathbb{Z}$; Theorem 10 therefore places $(p/q,r/q)$ in $\mathcal{S}_\infty$.

Finally, $\mathcal{S}_1 = \mathcal{W} \setminus \mathcal{S}_\infty$. Indeed, a point of $\mathcal{W}$ with irrational α has exactly one hit by Theorem 12, whereas a point of $\mathcal{W}$ with rational α is automatically compatible and hence recurrent by Theorem 10. Thus $\mathcal{S}_1$ has measure zero. It is uncountable, and it is dense because through every recurrent point passes a web segment containing irrational-α points arbitrarily nearby; the density of $\mathcal{S}_\infty$ then gives the claim.

The map of Theorem 6 is a smooth bijection with smooth inverse on the two open interiors. It therefore preserves density and cardinality, and, locally on compact subsets, preserves null sets in both directions. The same stratification statements consequently hold in $\mathcal{P}$. □

Further exact spectral consequences. The next two consequences are not needed for conditioning, but they complete the fixed-level spectral picture. Write $n_+(A)$ for the number of positive eigenvalues of a real symmetric matrix A, counted with multiplicity. Quantization makes the inertia immediate: a shifted eigenvalue is positive exactly when $\theta_k < \theta_\tau$.

Proposition 14 (Inertia staircase)**.** *For every interior parameter pair and $N \geq 2$,*

$$\boxed{n_+(B_N(a,\tau)) = \lceil N\alpha - \beta \rceil\,.} \tag{41}$$

If $N\alpha - \beta \in \mathbb{Z}$, exactly one eigenvalue is zero; otherwise none is zero. The number of negative eigenvalues is the remainder.

Proof. By monotonicity of μ_a, $\lambda_k > 0$ iff $\theta_k < \theta_\tau$. Since G_N is increasing, this is equivalent to

$$k\pi < G_N(\theta_\tau) = \pi(N\alpha + 1 - \beta).$$

Counting integers $1 \leq k \leq N$ below this threshold gives $\lceil N\alpha - \beta \rceil$. Equality in the threshold is exactly the singularity criterion. □

Proposition 15 (Kernel)**.** *At a singular dimension,*

$$\ker B_N(a,\tau) = \operatorname{span}\left\{ \big(\sin((j+1)\theta_\tau) - a\sin(j\theta_\tau)\big)_{j=0}^{N-1} \right\}. \tag{42}$$

Proof. Evaluate the eigenvector formula (22) at the crossing angle, which is an eigenangle precisely at singularity. Nullity one was proved in Theorem 9. □

5. Conditioning across the symbol band

The three positions relative to the symbol band now produce three local mechanisms. At an interior crossing the symbol has nonzero slope, so phase error converts linearly into spectral error. At either edge that slope vanishes and the first nonzero term is quadratic. Outside the band there is no crossing and a fixed gap remains. We treat the interior first, then the exterior and edge cases.

All interior asymptotic statements in this section are for a fixed pair (a,τ) with $\mu_- < \tau < \mu_+$; implied constants may depend on that pair. We first treat the interior band. Put

$$\gamma(a,\tau) = |\mu_a'(\theta_\tau)|. \tag{43}$$

A direct substitution into the derivative of (3) gives

$$\boxed{\gamma(a,\tau) = \frac{2a\tau^2 \sin(\pi\alpha)}{1-a^2}.} \tag{44}$$

Equivalently, differentiating $\mu(\theta) = \sin\psi_a(\theta)/\sin\theta$ and using $\psi_a' = \mu_a$ gives

$$\gamma(a,\tau) = \tau\big(\cot(\pi\alpha) - \tau\cot(\pi(1-\beta))\big), \tag{45}$$

with the sign understood through the absolute value in (43).

The interior proof has two steps. First locate the eigenangle whose phase index is nearest the fixed crossing; then pass from angular displacement to spectral displacement through the slope $\gamma(a,\tau)$.

For sufficiently large N, the nearest integer to $G_N(\theta_\tau)/\pi = N\alpha + 1 - \beta$ belongs to the physical index set $\{1, \dots, N\}$. Choose such a nearest integer k_N.

Theorem 16 (Nonasymptotic localization)**.** *For all sufficiently large N,*

$$\frac{\pi d_N}{N + \mu_+} \le |\theta_{k_N} - \theta_\tau| \le \frac{\pi d_N}{N + \mu_-}. \tag{46}$$

For nonsingular dimensions,

$$\boxed{|\theta_{k_N} - \theta_\tau| = \frac{\pi d_N}{N + \tau}\big(1 + O(N^{-2})\big).} \tag{47}$$

Proof. The distance from $G_N(\theta_\tau)$ to the nearest multiple of π is exactly πd_N. The mean-value theorem and $N + \mu_- \le G_N' \le N + \mu_+$ yield (46). Hence the intermediate point lies $O(N^{-1})$ from θ_τ, so $\mu_a(\xi) = \tau + O(N^{-1})$. Thus

$$N + \mu_a(\xi) = N + \tau + O(N^{-1}),$$

a relative $O(N^{-2})$ perturbation of $N + \tau$, giving (47). □

The preceding theorem is a phase-space estimate. Because the crossing is interior, $|\mu_a'(\theta_\tau)| = \gamma(a, \tau) > 0$, so it now translates directly into the smallest singular value.

Theorem 17 (Interior smallest singular value)**.** *For every fixed interior pair and every sufficiently large nonsingular dimension,*

$$\boxed{\sigma_{\min}(B_N(a, \tau)) = \pi\gamma(a, \tau)\frac{d_N}{N + \tau}\big(1 + O(N^{-1})\big).} \tag{48}$$

At an exact singular dimension, $\sigma_{\min} = 0$.

Proof. By monotonicity of μ_a, the eigenvalue nearest zero comes from one of the two secular roots bracketing θ_τ. Their phase displacements from $G_N(\theta_\tau)/\pi$ are

$$r_1 = d_N, \qquad r_2 = 1 - d_N,$$

with $0 < d_N \le 1/2$ at a nonsingular dimension. Applying the localization argument to each bracketing root (with r_j in place of d_N) gives, uniformly for $j = 1, 2$,

$$|\theta^{(j)} - \theta_\tau| = \frac{\pi r_j}{N + \tau}\big(1 + O(N^{-2})\big).$$

Taylor expansion at the interior crossing, where $\gamma = |\mu_a'(\theta_\tau)| > 0$, then gives

$$|\mu_a(\theta^{(j)}) - \tau| = \pi\gamma(a, \tau)\frac{r_j}{N + \tau}\big(1 + O(N^{-1})\big).$$

The error bound is uniform in $r_j \in (0, 1]$. Since $r_2 \ge r_1 = d_N$, the minimum of the two displayed quantities is

$$\pi\gamma(a, \tau)\frac{d_N}{N + \tau}\big(1 + O(N^{-1})\big),$$

even in the near-tie case $d_N \approx 1/2$. This proves (48). □

The extreme KMS eigenvalues approach the symbol edges with $O(N^{-2})$ error, so

$$\sigma_{\max}(B_N(a, \tau)) = M(a, \tau) + O(N^{-2}), \tag{49}$$

where

$$M(a, \tau) = \max\{\mu_+ - \tau, \tau - \mu_-\}. \tag{50}$$

Theorem 18 (Interior conditioning law)**.** *Define*

$$C(a,\tau)=\frac{M(a,\tau)}{\pi\gamma(a,\tau)}. \tag{51}$$

For every fixed interior pair and every sufficiently large nonsingular N,

$$\boxed{\kappa_2(B_N(a,\tau))=C(a,\tau)\frac{N+\tau}{d_N}\big(1+O(N^{-1})\big).} \tag{52}$$

Proof. Divide (49) by (48). □

This completes the interior mechanism. Outside the band the nearest spectral point stays a positive distance from τ; at an edge, by contrast, the interior slope vanishes. These two facts complete the remaining four regimes.

Theorem 19 (Outside the symbol band)**.** *If $\tau<\mu_-$, then*

$$\kappa_2(B_N(a,\tau))<\frac{\mu_+-\tau}{\mu_--\tau},\qquad \kappa_2(B_N(a,\tau))\longrightarrow\frac{\mu_+-\tau}{\mu_--\tau}. \tag{53}$$

If $\tau>\mu_+$, then

$$\kappa_2(B_N(a,\tau))<\frac{\tau-\mu_-}{\tau-\mu_+},\qquad \kappa_2(B_N(a,\tau))\longrightarrow\frac{\tau-\mu_-}{\tau-\mu_+}. \tag{54}$$

In particular the condition number is uniformly bounded in N outside the band.

Proof. Every finite KMS eigenvalue lies strictly between the symbol edges. Below the band all shifted eigenvalues are positive and above the band all are negative; the inequalities follow by taking the ratio of the farther and nearer edge distances. The limits follow from convergence of the extreme eigenvalues to $\mu_\pm$. □

At the band edges the crossing occurs where μ_a' vanishes, so the interior linearization is replaced by a quadratic Taylor term.

Theorem 20 (Band-edge conditioning)**.** *At the upper edge $\tau=\mu_+$,*

$$\sigma_{\min}(T_N(a)-\mu_+I_N)=\frac{\pi^2a(1+a)}{(1-a)^3}\frac{1}{(N+\mu_+)^2}\big(1+O(N^{-2})\big), \tag{55}$$

and

$$\boxed{\kappa_2(T_N(a)-\mu_+I_N)=\frac{4\mu_-^2}{\pi^2}(N+\mu_+)^2\big(1+O(N^{-2})\big).} \tag{56}$$

At the lower edge $\tau=\mu_-$,

$$\sigma_{\min}(T_N(a)-\mu_-I_N)=\frac{\pi^2a(1-a)}{(1+a)^3}\frac{1}{(N+\mu_-)^2}\big(1+O(N^{-2})\big), \tag{57}$$

and

$$\boxed{\kappa_2(T_N(a)-\mu_-I_N)=\frac{4\mu_+^2}{\pi^2}(N+\mu_-)^2\big(1+O(N^{-2})\big).} \tag{58}$$

Proof. At the upper edge the first secular root satisfies

$$\theta_1=\frac{\pi}{N+\mu_+}+O(N^{-4}),$$

because $\psi_a(\theta)=\mu_+\theta+O(\theta^3)$ near 0. Since

$$\mu_a''(0)=-\frac{2a(1+a)}{(1-a)^3},$$

Taylor expansion gives (55). The largest singular value tends to the band width $\mu_+ - \mu_- = 4a/(1-a^2)$, yielding (56). The lower edge is identical after writing $\varepsilon_N = \pi - \theta_N$; there

$$\varepsilon_N = \frac{\pi}{N+\mu_-} + O(N^{-4}), \qquad \mu_a''(\pi) = \frac{2a(1-a)}{(1+a)^3}.$$

This gives (57) and (58). □

Equations (52), (53), (54), (56) and (58) prove Theorem 1. Notice that the interior formula is necessarily nonuniform as $\tau \to \mu_\pm$, because $\gamma(a,\tau) = |\mu_a'(\theta_\tau)| \to 0$ at the band edges. The vanishing slope is exactly what converts the interior N^{-1} smallest singular-value scale into the edge N^{-2} scale.

6. Arithmetic laws for interior conditioning

All matrix-specific asymptotics are now contained in Theorem 18. What remains is the arithmetic of

$$d_N = \|N\alpha - \beta\|.$$

We separate three behaviors: periodic rational orbits, irrational bulk statistics, and rare inhomogeneous spikes.

6.1. Rational phase points.

Let $\alpha = p/q$ in lowest terms and define the q periodic distances

$$d_r = \left\| \frac{rp}{q} - \beta \right\|, \qquad r = 0, \dots, q-1. \tag{59}$$

Write δ_x for the unit point mass at x, and abbreviate $C = C(a,\tau)$ when no confusion can arise.

Theorem 21 (Rational normalized laws)**.** *If $q\beta \notin \mathbb{Z}$, then every $d_r > 0$ and, on $N \equiv r \pmod q$,*

$$\frac{\kappa_2(B_N)}{N} \longrightarrow \frac{C(a,\tau)}{d_r}. \tag{60}$$

Hence the empirical law of κ_2/N converges to the finite atomic measure

$$\frac{1}{q}\sum_{r=0}^{q-1} \delta_{C/d_r}. \tag{61}$$

If $q\beta \in \mathbb{Z}$, exactly one residue class has $d_r = 0$ and is singular; the other classes obey (60), so the limiting empirical law has one atom of mass $1/q$ at $+\infty$ and finite atoms on the remaining classes.

Proof. The sequence d_N is periodic mod q. Apply Theorem 18 on each nonsingular residue class and Theorem 10 on the compatible singular class. □

A rational-incompatible family. Take

$$\alpha = \frac{1}{3}, \qquad \beta = \frac{1}{2}.$$

Then $3\beta = 3/2 \notin \mathbb{Z}$, so no dimension is singular despite the rational crossing angle. The inverse map gives

$$a = 2 - \sqrt{3}, \qquad \tau = \frac{2}{\sqrt{3}}. \tag{62}$$

The distances are periodic:

$$d_N = \begin{cases} 1/2, & N \equiv 0 \pmod 3, \\ 1/6, & N \equiv 1, 2 \pmod 3. \end{cases}$$

Here $\gamma = 2/3$, $M = 1/\sqrt{3}$, and $C = \sqrt{3}/(2\pi)$, so

$$\frac{\kappa_2(B_N)}{N} \longrightarrow \begin{cases} \sqrt{3}/\pi, & N \equiv 0 \pmod 3, \\ 3\sqrt{3}/\pi, & N \equiv 1, 2 \pmod 3. \end{cases}$$

This branch cannot occur on the resonant line, where rational compatibility is automatic.

6.2. **Irrational phase points: bulk laws.** For irrational α, periodic residue classes disappear and equidistribution modulo one becomes the bulk mechanism. By Weyl's theorem [12], $\{N\alpha - \beta\}$ is equidistributed mod 1, independently of the target β. Hence d_N has the uniform empirical law on $[0, 1/2]$. We write $U[a, b]$ for the uniform probability law on $[a, b]$. A possible single exact singular dimension from Theorem 12 does not affect weak empirical limits.

Theorem 22 (Irrational empirical laws)**.** *For irrational α and $L \geq 1$, let*

$$\mathcal{I}_L = \{2, \dots, L+1\}, \qquad \mathcal{I}_L^* = \{N \in \mathcal{I}_L : d_N > 0\}, \qquad L^* = |\mathcal{I}_L^*|.$$

Then, weakly as $L \to \infty$,

$$\frac{1}{L} \sum_{N \in \mathcal{I}_L} \delta_{N\sigma_{\min}(B_N)} \Longrightarrow U\left[0, \frac{\pi\gamma(a,\tau)}{2}\right], \tag{63}$$

and

$$\frac{1}{L^*} \sum_{N \in \mathcal{I}_L^*} \delta_{\kappa_2(B_N)/N} \Longrightarrow \mathrm{Law}\left(\frac{C(a,\tau)}{U}\right), \qquad U \sim U[0, 1/2]. \tag{64}$$

Here $L^* = L$ except possibly for all sufficiently large L after one isolated singular dimension, in which case $L^* = L - 1$. The limiting condition-number law has Pareto tail index 1:

$$\Pr\left(\frac{C}{U} > t\right) = \min\left\{1, \frac{2C}{t}\right\}, \qquad (t > 0). \tag{65}$$

Moreover, after deleting any isolated singular term,

$$\frac{1}{L} \sum_{N \in \mathcal{I}_L^*} \frac{\kappa_2(B_N)}{N} \longrightarrow +\infty. \tag{66}$$

Proof. Equidistribution gives weak convergence of the empirical measures of d_N to $U[0, 1/2]$; deleting at most one index does not change the limit. The first two statements then follow from the continuous-mapping theorem and Theorems 17–18. For (66), truncate $x \mapsto C/x$ at height R, average the resulting bounded function by equidistribution, and then let $R \to \infty$; its integral against $U[0, 1/2]$ diverges logarithmically. □

6.3. **Exceptional spikes.** The bulk law does not control the rare dimensions for which d_N is much smaller than a typical equidistributed value. Those dimensions govern the largest condition-number spikes and require one-sided inhomogeneous approximation.

Following the one-sided inhomogeneous literature, define

$$\mathcal{M}_+(\alpha, \beta) = \liminf_{N \to \infty} N \|N\alpha - \beta\|. \tag{67}$$

See [15] for the corresponding one-sided spectrum.

Theorem 23 (Quadratic envelope)**.** *For irrational α, after deleting any isolated exact singular dimension,*

$$\boxed{\limsup_{N \to \infty} \frac{\kappa_2(B_N)}{N^2} = \frac{C(a,\tau)}{\mathcal{M}_+(\alpha,\beta)},} \tag{68}$$

with $C/0 = +\infty$.

Proof. By Theorem 18,

$$\frac{\kappa_2(B_N)}{N^2} = \frac{C(a,\tau)}{Nd_N}(1+o(1)),$$

and taking the limsup gives the reciprocal of the defining liminf (67). □

For logarithmic growth, it is useful to keep the positivity of the matrix dimension explicit.

Definition 24. Define the positive-direction inhomogeneous approximation exponent

$$w_+(\alpha,\beta) = \sup\left\{w \geq 0 : \|N\alpha - \beta\| < N^{-w} \text{ for infinitely many } N \in \mathbb{N}\right\}. \tag{69}$$

This is the one-sided analogue of the inhomogeneous exponents studied in the general framework of Bugeaud–Laurent [17].

Theorem 25 (Conditioning exponent)**.** *For irrational α, after deleting any isolated singular dimension,*

$$\boxed{\limsup_{N\to\infty} \frac{\log \kappa_2(B_N)}{\log N} = 1 + w_+(\alpha,\beta).} \tag{70}$$

Proof. Take logarithms in

$$\kappa_2(B_N) = C(a,\tau)(N+\tau)d_N^{-1}(1+o(1))$$

and compare the limsup of $\log(1/d_N)/\log N$ with Definition 24. □

Classical metric inhomogeneous approximation now yields a strong generic statement. Cassels' theorem implies, for fixed irrational α and every $c > 0$, that

$$\|N\alpha - \beta\| < \frac{c}{N} \tag{71}$$

has infinitely many positive solutions for almost every target β; see the summary and extensions in [16, 18]. Intersecting over $c = 1/j$ gives $\mathcal{M}_+(\alpha,\beta) = 0$ almost everywhere. On the other hand, for every $\varepsilon > 0$ the target intervals for $\|N\alpha - \beta\| < N^{-1-\varepsilon}$ have summable total length, so the first Borel–Cantelli lemma gives $w_+ \leq 1$ almost everywhere. Equation (71) with $c = 1$ gives the reverse inequality.

Theorem 26 (Generic shifted conditioning)**.** *Fix irrational α with $0 < \alpha < 1$. For Lebesgue-almost every target $0 < \beta < 1 - \alpha$,*

$$\mathcal{M}_+(\alpha,\beta) = 0, \qquad w_+(\alpha,\beta) = 1. \tag{72}$$

Consequently

$$\boxed{\limsup_{N\to\infty} \frac{\log \kappa_2(B_N)}{\log N} = 2, \qquad \limsup_{N\to\infty} \frac{\kappa_2(B_N)}{N^2} = +\infty.} \tag{73}$$

Thus a generic interior shift exhibits three scales at once: linear bulk behavior in distribution, quadratic growth on the logarithmic exceptional scale, and no finite quadratic limsup envelope.

7. The resonant hollow slice

The general shifted problem is now complete. The hollow case $\tau = 1$ is a distinguished line inside the same target triangle, not a separate spectral mechanism. Along that line the affine residual becomes homogeneous, so the earlier classification and conditioning laws specialize directly.

Theorem 27 (Resonance)**.** *Within the interior triangle,*

$$\boxed{\tau = 1 \iff \beta = \alpha.} \tag{74}$$

Moreover

$$\beta > \alpha \iff \tau > 1, \qquad \beta < \alpha \iff \tau < 1. \tag{75}$$

Proof. By (28), $\tau = 1$ iff $\sin(\pi\beta) = \sin(\pi\alpha)$. Since $\alpha, \beta > 0$ and $\alpha+\beta < 1$, the alternative $\beta = 1-\alpha$ is excluded; hence $\beta = \alpha$. The sign statement follows from

$$\sin(\pi\beta) - \sin(\pi\alpha) = 2\cos\frac{\pi(\alpha+\beta)}{2}\sin\frac{\pi(\beta-\alpha)}{2},$$

whose cosine factor is positive in the triangle. □

The singularity web is $\beta \equiv N\alpha \pmod 1$, so the resonant diagonal $\beta = \alpha$ is exactly its $N = 1$ member. Our matrix range starts at $N \geq 2$; if $N = 1$ is restored, then $T_N(a) = [1]$ and $B_N(a, 1) = [0]$. Thus the web interpretation extends literally to the omitted one-dimensional case.

Corollary 28 (Hollow rational/irrational dichotomy)**.** *At $\tau = 1$,*

$$d_N = \|(N-1)\alpha\|\,. \tag{76}$$

Hence, if $\alpha = p/q$ in lowest terms, $T_N(a) - I_N$ is singular exactly when

$$N \equiv 1 \pmod q, \tag{77}$$

whereas irrational α gives no singular physical dimensions.

Proof. On $\beta = \alpha$, the singularity criterion is $(N-1)\alpha \in \mathbb{Z}$ and (76) is immediate. □

The resonance is not confined to exact singularity. Every affine web line reduces the inhomogeneous problem to a homogeneous one after reindexing.

Proposition 29 (Affine-hit homogeneous collapse)**.** *If for some fixed integer $r \geq 1$,*

$$\beta \equiv r\alpha \pmod 1, \tag{78}$$

then

$$\boxed{d_N = \|(N-r)\alpha\|\,.} \tag{79}$$

Consequently

$$\mathcal{M}_+(\alpha, \beta) = \liminf_{q\to\infty} q\,\|q\alpha\|\,. \tag{80}$$

Proof. Equation (78) means $\beta = r\alpha - m$ for some integer m, so

$$\|N\alpha - \beta\| = \|(N-r)\alpha + m\| = \|(N-r)\alpha\|\,.$$

Replacing N by $q + r$ does not change the liminf after multiplication by $N/q \to 1$. □

An isolated irrational singularity. Take

$$a = \frac{1}{2}, \qquad \tau = \frac{3}{2}.$$

At $N = 2$,

$$T_N(1/2) - \frac{3}{2}I_2 = \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$$

is singular. The crossing angle satisfies $\cos\theta_\tau = 3/4$, so

$$\alpha = \frac{\arccos(3/4)}{\pi}$$

is irrational, while the target lies on the physical line $\beta = 2\alpha$. Theorem 12 therefore says that $N = 2$ is the only singular dimension. After that exact hit, Proposition 29 gives $d_N = \|(N-2)\alpha\|$.

Thus the classical homogeneous Markov–Lagrange theory reappears on every affine-hit line. In particular, on $\tau = 1$ it gives the Hurwitz bound and the familiar noble/golden extremality of the normalized arithmetic constant; see [13, 14]. Away from these affine reductions, the target β is a genuine second arithmetic coordinate: $\mathcal{M}_+(\alpha, \beta)$ depends on it, and for almost every target it vanishes.

The spectral constants also collapse exactly to those of the hollow matrix. On $\tau = 1$ one has $a = \cos(\pi\alpha)$,

$$\gamma(a,1) = \frac{2a}{\sqrt{1-a^2}}, \qquad M(a,1) = \frac{2a}{1-a}, \tag{81}$$

and therefore

$$C(a,1) = \frac{1}{\pi}\sqrt{\frac{1+a}{1-a}}. \tag{82}$$

Theorem 18 becomes

$$\kappa_2(T_N(a) - I_N) = \frac{1}{\pi}\sqrt{\frac{1+a}{1-a}}\,\frac{N+1}{\|(N-1)\alpha\|}\big(1 + O(N^{-1})\big). \tag{83}$$

The factor $N+1$ in the hollow slice is therefore just the specialization of the universal local factor $N+\tau$.

There is one further collapse at an exact hollow singularity. The general kernel vector (42) becomes, since $a = \cos\theta_\tau$,

$$\sin((j+1)\theta_\tau) - a\sin(j\theta_\tau) = \sin\theta_\tau\cos(j\theta_\tau).$$

Thus the two-sine nullvector reduces to the first-kind Chebyshev vector $(\cos(j\theta_\tau))_{j=0}^{N-1}$. Likewise the determinant (32) collapses to the single-Chebyshev formula for the hollow matrix. The homogeneous arithmetic, the Chebyshev kernel and the determinant collapse are three consequences of the same resonance.

8. Positioning, applications, and scope

We close by recording the numerical interpretation of the preceding results and by separating the paper's KMS-specific contribution from its classical inputs.

The matrix $T_N(a)$ is the finite AR(1) correlation matrix, a standard covariance/correlation model rather than an artificial test matrix. The shifted problem also has an intrinsic numerical meaning: because $T_N(a)$ is symmetric, $\sigma_{\min}(T_N(a) - \tau I)$ is exactly the distance from τ to its finite spectrum. An exact hit makes the shifted system singular; a near hit makes it ill-conditioned. The results above describe how this stability changes with dimension when τ is fixed. We therefore keep the application discussion at this matrix-analysis level rather than attach the results to a more specific downstream model.

The novelty boundary matters because all three ingredients–KMS spectral theory, phase quantization, and Diophantine approximation–have substantial prior literature. The subsections below separate those inputs from the KMS-specific conclusions.

Classical KMS and phase input. The KMS matrix, its tridiagonal inverse, Poisson-kernel symbol, sinusoidal eigenvectors and characteristic-polynomial descriptions are classical [1, 2, 4, 3]. In particular, Chremmos and Fikioris derive a linear second-order recurrence for the characteristic polynomial of the geometric Toeplitz family [4]; its real KMS specialization contains the hollow determinant identity. The secular equation and its phase form also sit naturally in the finite Toeplitz/simple-loop framework [7, 8]. Thus the Chebyshev formula (30) is a specialization and reorganization of known characteristic-polynomial machinery, not a new determinant technology. What matters here is its organization around a fixed level τ and the residual $N\alpha - \beta$.

Proposition 3 identifies the KMS boundary phase with the disk automorphism b_a. Boundary phases of Blaschke products and related Prüfer variables are established tools for spectral and orthogonal-polynomial zero problems [21]; exact finite-Blaschke quantization also occurs in finite Bernstein–Szegő Jacobi systems [22]. We therefore do not claim the Blaschke/phase mechanism itself as new. Its useful KMS specialization is the identity $\psi_a' = \mu_a$: at a fixed level crossing it makes $G_N'(\theta_\tau) = N + \tau$, so the local conditioning constant is especially explicit.

Diophantine spectral proximity and Toeplitz conditioning. The general observation that finite-matrix spectral proximity can be controlled by Diophantine properties of an angle has direct precedent. Davies–Levitin reduce a spectral-distance problem to the nearest point in a cosine mesh and explicitly state that the rate of approach to zero is determined by the Diophantine properties of $\pi^{-1}\arccos(c/2)$ [19]. Thus the mere appearance of Diophantine approximation in a matrix spectrum is not a novelty claim of this paper.

Toeplitz conditioning has likewise long been tied to the zeros and winding of the symbol [10, 11, 20]. Of particular relevance, Bogoya–Böttcher–Ferrari–Grudsky–Serra-Capizzano give, for a tridiagonal Toeplitz example with two conjugate unit-circle zeros, an exact inverse norm equal to the reciprocal of the nearest cosine-mesh distance, and note that the resulting condition numbers may diverge irregularly [9]. Accordingly, neither irregular mesh-driven conditioning nor the broad bounded versus polynomially unbounded regime structure is claimed as new here.

The problem here is more specific: fix the real parameter a and the real level τ, then vary the dimension. The exact KMS phase yields not only a nearest-mesh description but the affine inhomogeneous residual

$$d_N = \|N\alpha - \beta\|$$

together with an explicit parameter map, an exact determinant, and the leading smallest-singular-value constant. This makes it possible to follow the same reduction through rational compatibility, irrational empirical laws, and one-sided inhomogeneous exceptional behavior.

The complex-parameter KMS literature studies a different prescribed-level problem: Fikioris–Mavrogordatos treat borderline and extraordinary eigenvalues, while Fikioris–Papapanos construct complex-ρ level curves on which a fixed $K_n(\rho)$ has an eigenvalue of prescribed magnitude [5, 6]. There the dimension is fixed and the matrix parameter varies; here (a, τ) is fixed and recurrence is studied across dimensions.

Imported arithmetic input. The arithmetic results in Section 6 are applications of established inhomogeneous Diophantine approximation. We use the one-sided inhomogeneous constant and spectrum studied by Crisp–Moran–Pollington [15], the metric theorem of Cassels [18] as summarized and extended in [16], and the inhomogeneous exponent framework of Bugeaud–Laurent [17]. These are not new number-theoretic results. The contribution is their exact translation into this shifted-KMS conditioning problem.

What is claimed here. Subject to the literature boundary above, the paper establishes the following KMS-specific package: explicit target-triangle coordinates (28); the affine fixed-level singularity web (36); the $0/1/\infty$ classification with complete rational compatibility parametrization; the interpretation of $\tau = 1$ as the $N = 1$ member of that web; the exact five-regime conditioning phase diagram with explicit constants; rational compatible/incompatible normalized laws and the irrational Pareto bulk law; and the reduction of exceptional conditioning to the one-sided inhomogeneous constant $\mathcal{M}_+(\alpha, \beta)$.

Within the KMS, Toeplitz-conditioning, and Diophantine literature surveyed here, we have not found this full fixed-level scalar-shift package. The claim is not that every algebraic reformulation is new: the spectral ingredients are classical, there are clear predecessors for Diophantine spectral

proximity, and the arithmetic theorems are imported. The novelty claim is the exact KMS-specific assembly and quantitative dictionary.

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Open directions. First, the one-sided inhomogeneous spectrum $\mathcal{S}_+(\alpha)$ is richer than the homogeneous Markov–Lagrange spectrum. A sharper study of which positive values of $\mathcal{M}_+(\alpha,\beta)$ occur inside the restricted KMS target interval $0 < \beta < 1-\alpha$ could refine the exceptional conditioning picture. Second, the degree-one Blaschke identity suggests a broader exact-phase transfer principle: finite Bernstein–Szegő Jacobi systems already provide a genuinely different exact finite-Blaschke family [22]. We leave that abstract generalization separate so that the present paper remains a complete KMS-specific result. Finally, non-real KMS parameters would require singular-value rather than Hermitian eigenvalue control and are not treated here.

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